

PUNJAB RETROFITTING ROADS

Sector: Infrastructure
Hazard addressed: Flood
Targeted assets: Roads



Description: This measure involves retrofitting existing roads in flood-prone districts of Punjab to improve their resilience against flood hazards. Interventions include raising road embankments above the 1-in-100-year flood level, construction of retaining walls, and installation of additional culverts with enhanced flow capacity to allow floodwater to pass without road damage.

Priority Districts: Mianwali, Chakwal, Gujrat, Gujranwala, Bahawalpur, Rahim Yar Khan, Lahore

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Per km of road.

Scale assumptions: Roads located in flood prone areas of the target districts.

Key design parameters: Raising road embankments above 1-in-100-year flood level; construction of retaining walls; installation of additional culverts with enhanced flow capacity.

Time horizon: 2026–2035

IMPACT

Risk reduction level: Very High Effectiveness – large reduction in damages (Expert Rank 1).

Time to effectiveness: Effectiveness begins progressively as individual road sections are retrofitted over 2026–2035, with full effectiveness from 2036 onwards.

Expected impact: 90% reduction in road damage (MDD = 0.1); protects road asset base valued at USD 418,727,992 across seven target districts.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Road asset inventory and condition survey; hydraulic design for embankment raising; procurement of qualified and experienced contractors; traffic management plans during retrofits/rehabilitation.

Required institutions: Punjab Communications & Works (C&W) Department; National Highway Authority (NHA); District governments and Traffic Police.

Barriers: Procurement complexity for large-scale road works; traffic disruption during construction; coordination gaps between Punjab C&W, NHA and district governments/police; budget release timing; access to flood-prone road sections during construction.

COST ASSUMPTIONS

Unit cost: USD 13,278 per km.

Initial Investment Cost (CAPEX):
USD 83,745,598.

Annual O&M costs (OPEX):
USD 133 per km per year.

CO-BENEFITS & TRADE-OFFS

Environmental: Properly identified and designed culverts allow natural water flow across roads without damming.

Social: Maintained road connectivity during floods enables emergency response and evacuation; continued market access for rural communities; reduced isolation of flood-affected areas.

Economic: Avoided road reconstruction costs damaged due to floods; maintained supply chains and freight movement; reduced indirect economic losses from transport disruption.

Potential risks: Traffic disruption during retrofitting works; access constraints in seasonally flooded areas; residual risk for events exceeding 1-in-100-year threshold; road maintenance quality affects long-term effectiveness.

BENEFICIARIES

Primary beneficiaries: Communities dependent on road connectivity for livelihoods, access to education and health facilities; emergency response agencies.

Distributional aspects: Particularly beneficial for rural and agricultural communities in flood-prone areas of target districts with limited alternative transport routes.